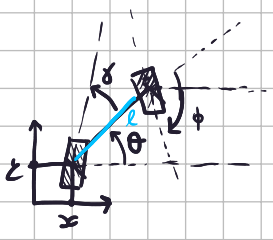
1)



$q = (x \; z \; \theta \; \phi \; \gamma)^T \in \mathbb{R}^2 \times SO(2)^3$ there are two kinematic constr.

The rear wheel must go towards $\theta + \gamma$:

$$(\dot{x} \;\; \dot{z}) \begin{pmatrix} -\sin(\theta+\gamma) \\ \cos(\theta+\gamma) \end{pmatrix} = 0 \Rightarrow -s_{\theta\gamma}\dot{x} + c_{\theta\gamma}\dot{z} = 0$$

The other constr. is on the Front wheel:

$$\begin{cases} x_s = x + \ell c_\theta \\ z_s = z + \ell s_\theta \end{cases} \Rightarrow \begin{cases} \dot{x}_s = \dot{x} - \ell s_\theta \dot{\theta} \\ \dot{z}_s = \dot{z} + \ell c_\theta \dot{\theta} \end{cases}$$

Must go toward $\theta + \phi$: $(\dot{x}_s \;\; \dot{z}_s) \begin{pmatrix} -\sin(\theta+\phi) \\ \cos(\theta+\phi) \end{pmatrix} = -s_{\theta\phi}\dot{x} + \ell s_\theta s_{\theta\phi}\dot{\theta} + c_{\theta\phi}\dot{z} + \ell c_\theta c_{\theta\phi}\dot{\theta} = 0$

$$\Rightarrow -s_{\theta\phi}\dot{x} + c_{\theta\phi}\dot{z} + \ell\dot{\theta}c_\phi$$

Let's write the constraints in matrix form:

$$\underbrace{\begin{pmatrix} -s_{\theta\gamma} & c_{\theta\gamma} & 0 & 0 & 0 \\ -s_{\theta\phi} & c_{\theta\phi} & \ell c_\phi & 0 & 0 \end{pmatrix}}_{A} \dot{q} = 0$$

let' look for a basis of $NS(A)$:

$g_3 = (0 \; 0 \; 0 \; 0 \; 1)^T$

$g_2 = (0 \; 0 \; 0 \; 1 \; 0)$ $g_1 = (c_{\theta\gamma} \;\; s_{\theta\gamma} \;\; \alpha \; 0 \; 0)^T$ where α should satisfy

$$\underbrace{-s_{\theta\phi}c_{\theta\gamma} + c_{\theta\phi}s_{\theta\gamma}}_{\sin(\gamma-\phi)} + \alpha\ell c_\phi = 0 \Rightarrow \alpha\ell c_\phi = \sin(\phi-\gamma) \Rightarrow \alpha = \frac{\sin(\phi-\gamma)}{\ell\cos\phi}$$

$\Rightarrow$ the model is $\dot{q} = g_1 v + g_2 \omega \Rightarrow$

$$\begin{cases} \dot{x} : v\cos(\theta+\gamma) \\ \dot{z} : v\sin(\theta+\gamma) \\ \dot{\theta} : v\,\frac{\sin(\phi-\gamma)}{\ell\cos\phi} \\ \dot{\phi} : \omega_1 \\ \dot{\gamma} : \omega_2 \end{cases} = Gu, \quad g_1 = \begin{pmatrix} c_{\theta\gamma} \\ s_{\theta\gamma} \\ \frac{\sin(\phi-\gamma)}{\ell\cos\phi} \\ 0 \\ 0 \end{pmatrix} \quad g_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad g_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Now we prove controllability, we compute:

$[g_1, g_2]$ $[g_1, g_3]$ $[g_2, g_3]$ notice how $\frac{\partial g_2}{\partial q} = \frac{\partial g_3}{\partial q} = 0$ (zero matrix)

$$g_4 = [g_1, g_2] = \frac{\partial g_2}{\partial q} g_1 - \frac{\partial g_1}{\partial q} g_2 = -\frac{\partial g_1}{\partial q} g_2$$

since $g_2 = (0 \; 0 \; 0 \; 1 \; 0)^T$, we compute the 4° column of $\frac{\partial g_1}{\partial q}$

$$\left(\frac{\partial g_1}{\partial q}\right)^4 = \frac{\partial g_1}{\partial \phi} = \left(0 \quad 0 \quad \frac{\cos(\phi-\gamma)}{\ell\cos\phi} - \frac{\sin(\phi-\gamma)}{\ell\sin\phi} \quad 0 \quad 0\right)^T$$

$$\Rightarrow [g_1 \;\; g_2] = -\left(0 \quad 0 \quad \frac{\cos(\phi-\gamma)}{\ell\cos\phi} - \frac{\sin(\phi-\gamma)}{\ell\sin\phi} \quad 0 \quad 0\right)^T$$

$$g_5 = [g_1, g_3] = -\frac{\partial g_1}{\partial q} g_3 = -\left(\frac{\partial g_1}{\partial \gamma}\right) = \begin{pmatrix} -S_{\theta\gamma} & C_{\theta\gamma} & -\frac{\cos(\phi-\gamma)}{\ell\cos\phi} & 0 & 0 \end{pmatrix}^T$$

$\text{rank}\begin{pmatrix} g_1 & g_2 & g_3 & g_4 & g_5 \end{pmatrix} = 5 \Rightarrow$ the model is controllable.

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2) Let's write the model:

$$\begin{cases} \dot{x} = v\cos(\theta+\gamma) \\ \dot{y} = v\sin(\theta+\gamma) \\ \dot{\theta} = v\frac{\sin(\phi-\gamma)}{\ell\cos\phi} \\ \dot{\phi} = \omega_1 \\ \dot{\gamma} = \omega_2 \end{cases}$$

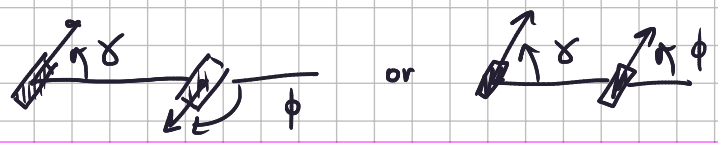
Let's try input - output linearization, we consider:

$$v = a\frac{\ell\cos\phi}{\sin(\phi-\gamma)} \Rightarrow \dot{\theta} = a \quad \text{then} \quad a = \dot{\theta}_d + K(\theta - \theta_d) \quad \text{with } K > 0$$

$$\Rightarrow \dot{\theta} = \dot{\theta}_d + K(\theta - \theta_d)$$

Feed Forward ↑ Feedback ↑

The controller achieves it's objective when $\sin(\phi-\gamma) \neq 0 \Rightarrow$ $\phi - \gamma \notin \{0, \pi\}$, in that situation there is a singularity.



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3) We will use the extended Kalman filter, we write the model at discrete time, with sampling interval T_s:

$$q_{k+1} = f(q_k, u_k) + n_k \Rightarrow \begin{cases} x_{k+1} = x_k + T_s v_k \cos(\theta_k + \gamma_k) + n_{k1} \\ y_{k+1} = y_k + T_s v_k \sin(\theta_k + \gamma_k) + n_{k2} \\ \theta_{k+1} = \theta_k + T_s v_k \frac{\sin(\phi_k - \gamma_k)}{\ell\cos\phi_k} + n_{k3} \\ \phi_{k+1} = \phi_k + T_s \omega_1 + n_{k4} \\ \gamma_{k+1} = \gamma_k + T_s \omega_2 + n_{k5} \end{cases}$$

These are the equations that we will use for prediction, using Euler integration for update.

The inputs u_k are constants in the sampling interval and $n_k \sim N(0, V_k)$.

We know that, in the EKF, if $\hat{q}_k$ is our current estimation, we predict a first estimate as $\hat{q}_{k+1|k} = f(\hat{q}_k, u_k)$. Then we need to make a correction using the measurements. We denote P_k the covariance matrix of the estimation, a prediction for P_k uses $F_k = \frac{\partial f}{\partial q}\Big|_{q=\hat{q}_k, u=u_k}$.

Before proceeding, we need to linearize the process dynamics. Let's compute $F_k = (F_{k1}^T \cdots F_{ns}^T)^T$:

$$F_k = \begin{bmatrix} 1 & 0 & -T_s v_k \sin(\hat{\theta}_k + \hat{\alpha}_k) & 0 & -T_s v_k \sin(\hat{\theta}_k + \hat{\alpha}_k) \\ 0 & 1 & T_s v_k \cos(\hat{\theta}_k + \hat{\alpha}_k) & 0 & T_s v_k \cos(\hat{\theta}_k + \hat{\alpha}_k) \\ 0 & 0 & 1 & \frac{T_s v_k}{\ell}\left(\frac{\cos(\hat{\theta}_k - \hat{\alpha}_k)}{\cos\hat{\phi}_k} - \frac{\sin(\hat{\phi}_k - \hat{\alpha}_k)}{\sin\hat{\phi}_k}\right) & -T_s v_k \frac{\cos(\hat{\theta}_k - \hat{\alpha}_k)}{\ell \cos\hat{\phi}_k} \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

A first estimate for the covariance matrix is obtained as $P_{k+1|k} = F_k P_k F_k^T + V_k$.

Now the correction phase, the charging station is at position $(x_L \; y_L)^T$ hence

$$z_{k+1} = h(q_{k+1}) = \sqrt{(x_{k+1} - x_L)^2 + (y_{k+1} - y_L)^2}$$

the innovation is $\nu_{k+1} = z_{k+1} - h(\hat{q}_{k+1|k})$

The prediction $\hat{q}_{k+1|k}$ is corrected as: $\hat{q}_{k+1} = \hat{q}_{k+1|k} + R_{k+1}\nu_{k+1}$ with R_{k+1} being the Kalman gain matrix.